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Course: Selected Topics in AI
Assignment: Proposal 12: Watershed Delineation and Stream Network Extraction from DEM

Watershed Delineation and Stream Network Extraction using MERIT Hydro and HydroSHEDS in Google Earth Engine

1. Objective

The primary objective of this project was to delineate a watershed and extract its associated stream network from elevation data, leveraging hydrological modeling techniques within the Google Earth Engine (GEE) platform. This involved understanding and implementing fundamental hydrological concepts such as flow accumulation and catchment boundary identification to visualize how terrain influences water movement and to explore the automated delineation of hydrological basins.

2. Overview of Methodology

This project followed the proposed methodology to perform hydrological analysis. It utilized Digital Elevation Model (DEM) data and pre-computed hydrological products to identify drainage patterns, delineate a specific watershed, and extract the corresponding stream network. The key steps involved were:

1. **Pour Point Definition:** A specific geographical point was defined, representing the outlet of the desired watershed.
2. **Data Acquisition:** The MERIT Hydro dataset was used for its upstream area (UPA) and elevation (HND) information, providing crucial hydrological parameters.
3. **Stream Network Identification:** A threshold on flow accumulation (represented by UPA in MERIT Hydro) was applied to distinguish significant stream channels from general land surface.
4. **Point Snapping:** To ensure the pour point accurately aligned with a significant stream, a mechanism was implemented to adjust the user-defined pour point to the nearest location with substantial upstream accumulation within a defined radius.

5. **Watershed Delineation:** The HydroSHEDS Level 7 (HYBAS) basin dataset was used to identify the pre-defined hydrological basin containing the snapped pour point, serving as the watershed boundary.
6. **Visualization and Analysis:** The resulting stream network and delineated watershed were visualized over a base map, and key statistics such as basin area and total stream area were computed.

3. Data Sources

The following datasets from the Google Earth Engine Data Catalog were utilized:

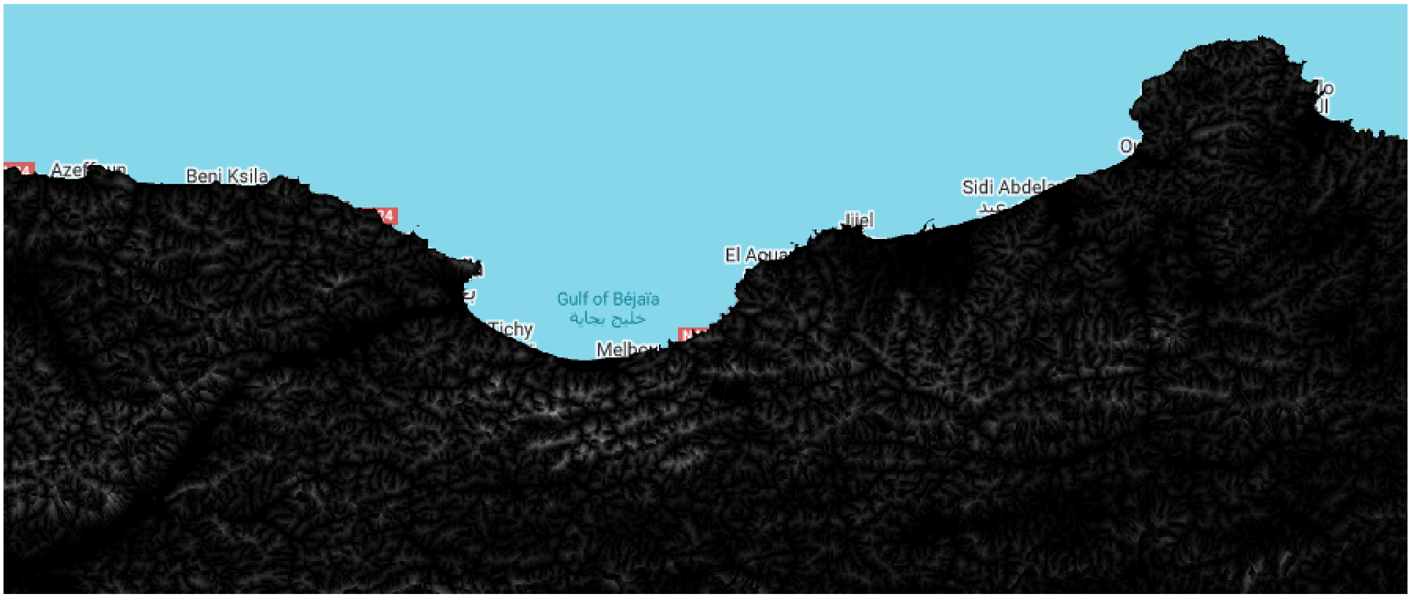
- **MERIT Hydro (v1_0_1):** This dataset provides high-resolution global hydrographic data, including Upstream Area (UPA), crucial for identifying stream networks, and Height above Nearest Drainage (HND) for elevation context.
- **WWF/HydroSHEDS/v1/Basins/hybas_7:** This dataset provides global hydrological basins at a nominal level 7, which were used for direct watershed delineation based on the snapped pour point.

4. Implementation and Visual Results

The analysis was performed entirely within the Google Earth Engine (GEE) code editor. The process involved several distinct visualization steps, which are presented below through screenshots of the GEE map interface.

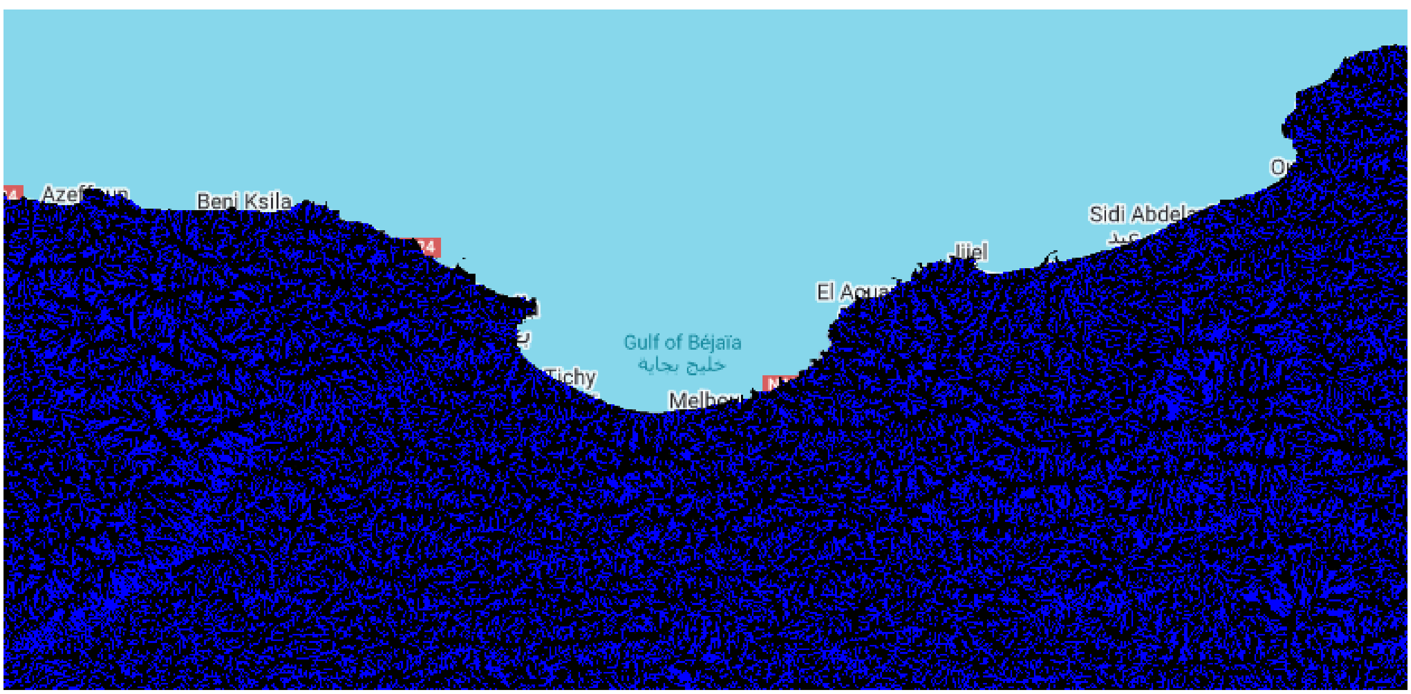
4.1. Initial Pour Point and MERIT Hydro Elevation/Upstream Area

The first step involved defining a pour point (the outlet of our target watershed) and loading the base hydrological data from MERIT Hydro. The MERIT elevation (HND) provides the underlying topography, while the Upstream Area (UPA) layer shows how much area contributes flow to each pixel. Darker blues in the UPA layer indicate higher accumulation.



4.2. Stream Network Identification

Based on a defined threshold for upstream area (0.05 km² in this case), a mask was created to highlight only the significant stream channels. This effectively filters out minor trickles and emphasizes the established drainage network.



4.3. Snapped Pour Point

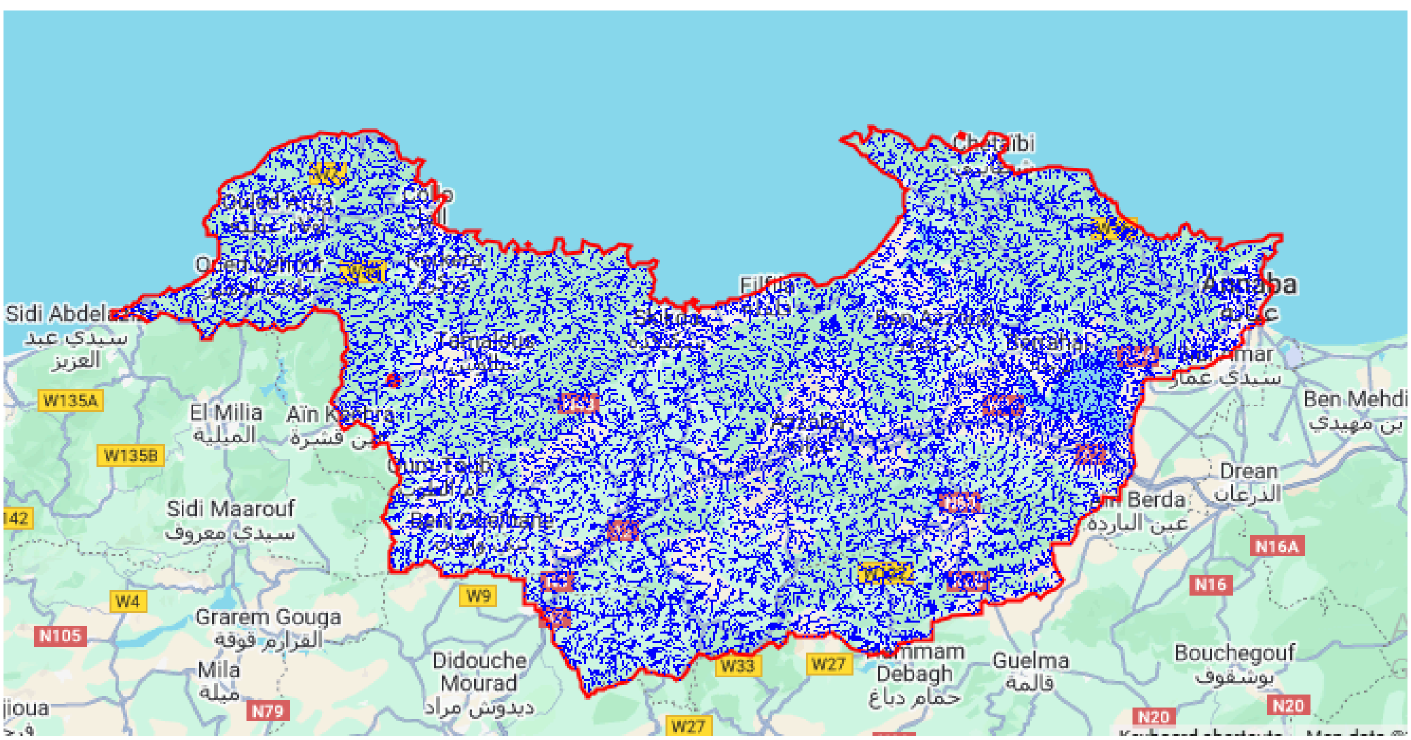
A critical step for accurate watershed delineation is to ensure the pour point is precisely located on a stream. The script automatically 'snapped' the initial pour point to the nearest significant stream segment within a 1000-meter radius, identifying the point of highest flow accumulation in that vicinity. This corrected pour point is marked in red.

4.4. Hydrological Basin Delineation (HYBAS Basin)

Using the snapped pour point, a pre-defined hydrological basin from the HydroSHEDS Level 7 dataset was identified. This basin outlines the entire catchment area that drains to the snapped pour point. The basin boundary is shown in red.

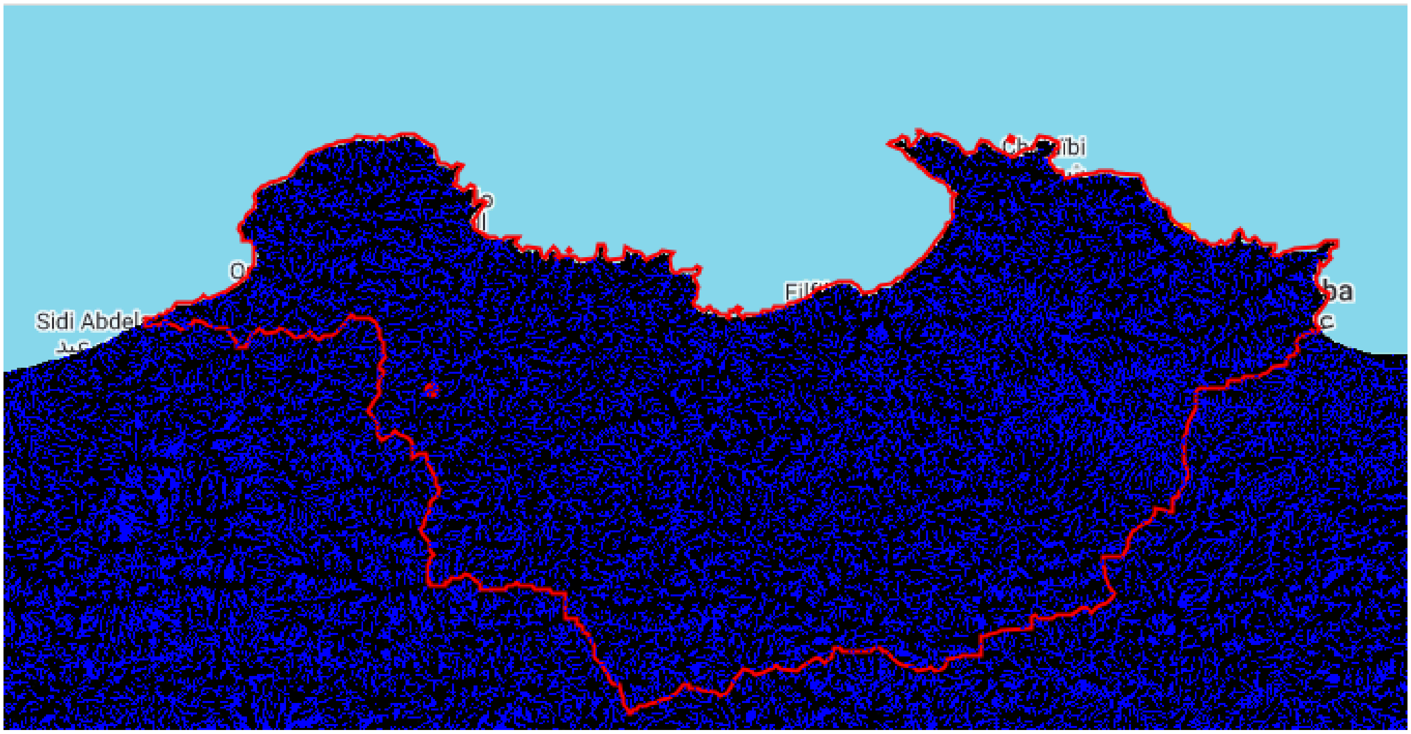
4.5. Clipped Stream Network within the Basin

To focus specifically on the drainage within our delineated watershed, the previously identified stream network was clipped to the boundaries of the HYBAS basin. This layer shows only the streams that contribute to the selected pour point.



4.6. Final Visualization

A combined view of the delineated watershed and its internal stream network, providing a comprehensive visual understanding of the hydrological system.



5. Quantitative Results

The GEE script also computed several key quantitative metrics, which are displayed in the GEE console. These metrics provide valuable numerical insights into the delineated watershed and its stream network.

**[Insert Screenshot Here: A screenshot of your GEE Console output, showing the print statements for:

- 'Max upa in buffer (km²):'
- 'Basin area (km²):'
- 'Stream area (m²):'
- 'Accumulation stats (km²):'
  Ensure all these values are visible in your console screenshot.]**

These values are:

- **Max UPA in buffer:** [Your actual value from GEE console] km²
- **Basin Area:** [Your actual value from GEE console] km²
- **Stream Area:** [Your actual value from GEE console] m²
- **Accumulation Statistics (Min, Max, Mean):** [Your actual values from GEE console] km²

6. Interpretation and Conclusion

The successful execution of this analysis in Google Earth Engine demonstrates several key aspects of hydrological modeling:

- **Terrain Dictates Water Movement:** The visualizations clearly show how the underlying elevation data directly influences the formation and path of stream networks and defines the boundaries of hydrological basins. Water naturally flows from higher to lower elevations, accumulating to form channels.
- **Automated Delineation:** The project successfully automated the delineation of a catchment area and extraction of its stream network from topographic data. This capability is vital for efficient large-scale or repeated hydrological analyses.
- **Support for Water Resource Management:** The outputs of this project – a clearly defined watershed and its internal stream network, along with quantitative metrics – are foundational for various applications in water resource management. For instance, knowing the basin area and stream extent is crucial for:
    - **Flood Modeling:** Predicting areas susceptible to flooding based on upstream contributions and flow paths.
    - **Water Supply Management:** Assessing the amount of water available within a catchment for human consumption, agriculture, or industry.
    - **Pollution Control:** Tracing potential sources of pollution within a watershed and understanding their impact on downstream areas.
    - **Infrastructure Planning:** Guiding the placement of dams, reservoirs, or other water management structures.